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EXPERIMENT 1

EXPERIMENT 1.1

Aim: Write a problem statement for your IPD project.

Title: Plant Disease Detection

Problem Statement:

Diseases in plants cause a threat to the yield and are a significant proportion why the global yield has reduced. Many a times, farmers themselves are not well educated in this field and hence, even after knowing that the crop has been affected with some disease, there is not much they can do to save the crop. Traditional methods are not very accurate and may also take a lot of time.

Plant disease detection application is a Machine Learning based application which detects common crop diseases. Users have to click a photo of the affected part of the plant and upload it to the app. Based on the uploaded image, the app will give the possible diagnosis based on the symptoms visible in the image and even provide treatment options. The application will provide a detailed report on the detected disease and available treatment options in an understandable format even to people who do not have knowledge about plant diseases. There will also be an e-commerce section in the app where users can shop for fertilizers, medicines and other products required for maintaining plant health. The app will have multiple language options to support diverse users.

Plagiarism Report:

English
French
Spanish
German

Problem Statement:

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AI-generated & AI-refined

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198 Words

Analysis complete

EXPERIMENT 1.2

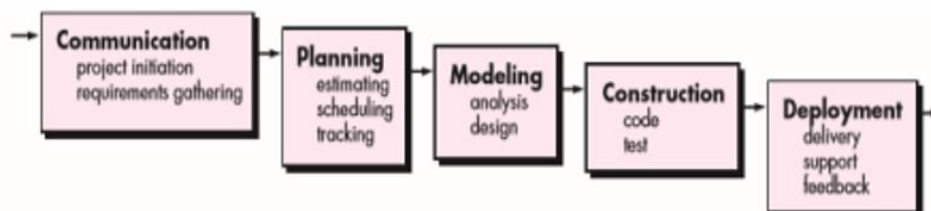
Aim: Describe various process models with diagrams.

Process Models:

1) Waterfall Model

The Waterfall Model is one of the most traditional approaches to project management, particularly in software development. It is a linear and sequential process, meaning that each phase of the project must be completed before moving to the next one. The stages typically include requirement gathering, system design, implementation, testing, deployment, and maintenance. Once a phase is finished, there is no going back, making this model suitable for projects with clearly defined requirements that are unlikely to change. However, it's less flexible when dealing with evolving needs and can lead to challenges if issues arise late in the project lifecycle.

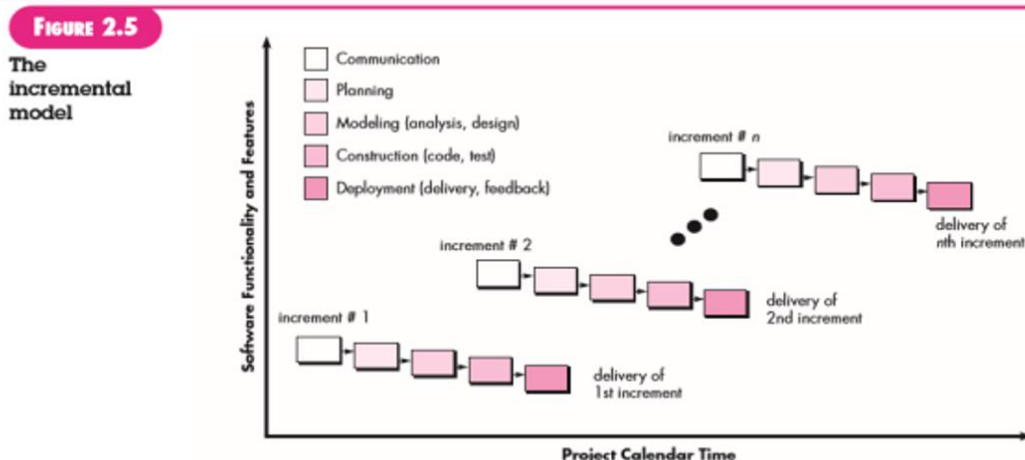
FIGURE 2.3 The waterfall model



- Advantages of waterfall model-
 - This model works for small projects because the requirements are understood very well.
 - The waterfall model is simple and easy to understand, implement, and use.
 - All the requirements are known at the beginning of the project, hence it is easy to manage.
- Disadvantages of the waterfall model
 - The problems with this model are uncovered, until the software testing.
 - The amount of risk is high.
 - This model is not good for complex and object oriented projects.

2) Incremental Process Model

The incremental model combines elements of linear and parallel process flows. Referring to Figure 2.5, the incremental model applies linear sequences in a staggered fashion as calendar time progresses. Each linear sequence produces deliverable “increments” of the software. When an incremental model is used, the first increment is often a core product. That is, basic requirements are addressed but many supplementary features (some known, others unknown) remain undelivered. The core product is used by the customer (or undergoes detailed evaluation). As a result of use and/or evaluation, a plan is developed for the next increment. The plan addresses the modification of the core product to better meet the needs of the customer and the delivery of additional features and functionality. This process is repeated following the delivery of each increment, until the complete product is produced. The incremental process model focuses on the delivery of an operational product with each increment. Early increments are stripped-down versions of the final product, but they do provide capability that serves the user and also provide a platform for evaluation by the user.



- Advantages of incremental model
 - This model is flexible because the cost of development is low and initial product delivery is faster.
 - It is easier to test and debug during the smaller iteration.
 - The working software generates quickly and early during the software life cycle.
 - The customers can respond to its functionalities after every increment.
- Disadvantages of the incremental model
 - The cost of the final product may cross the cost estimated initially.
 - This model requires a very clear and complete planning.
 - The planning of design is required before the whole system is broken into small increments.
 - The demands of customer for the additional functionalities after every increment causes problem during the system architecture.

3) Evolutionary Model

It is also called successive versions model or incremental model.

At first, a simple working model is built.

Subsequently it undergoes functional improvements & we keep on adding new functions till the desired system is built.

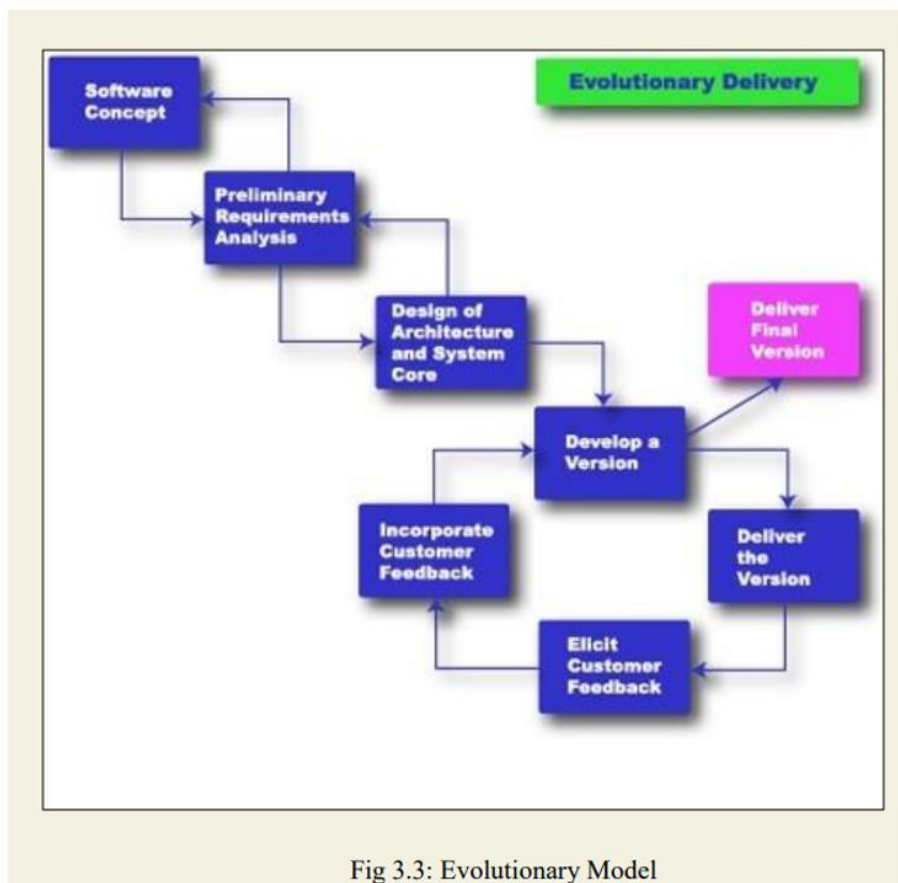


Fig 3.3: Evolutionary Model

- Advantages:
 - User gets a chance to experiment partially developed system.
 - Reduce the error because the core modules get tested thoroughly.

- Disadvantages:
 - It is difficult to divide the problem into several versions that would be acceptable to the customer which can be incrementally implemented & delivered.

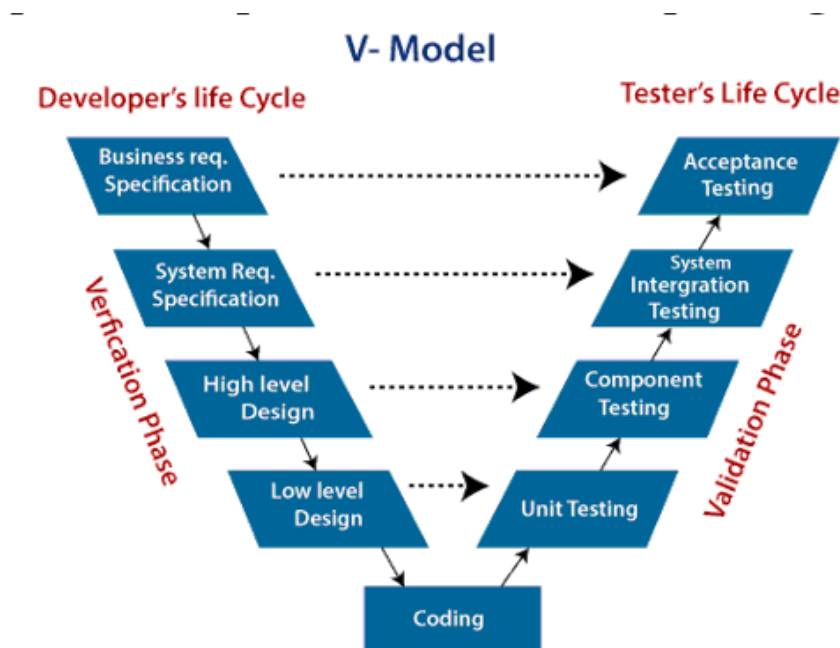
4) V Model

V-Model also referred to as the Verification and Validation Model.

In this, each phase of SDLC must complete before the next phase starts. It follows a sequential design process same as the waterfall model.

Testing of the device is planned in parallel with a corresponding stage of development.

- **Verification:** It involves a static analysis method (review) done without executing code. It is the process of evaluation of the product development process to find whether specified requirements meet.
- **Validation:** It involves dynamic analysis method (functional, non-functional), testing is done by executing code. Validation is the process to classify the software after the completion of the development process to determine whether the software meets the customer expectations and requirements.
- So V-Model contains Verification phases on one side of the Validation phases on the other side. Verification and Validation process is joined by coding phase in V-shape.
- Thus it is known as V-Model.



5) Concurrent Model

Concurrent engineering is an approach to product development in which multiple phases of the product development process are carried out simultaneously instead of sequentially. The main objective of concurrent engineering is to improve product quality and shorten development time by overlapping the different product life cycle stages. Concurrent engineering uses tools like development models to maximize the product development Process. Concurrent engineering is a holistic approach to product

development in which all stakeholders work together from the beginning to the end of the project.

Concurrent engineering is like a team sport because it requires close collaboration and communication among all members of the development team and the customer.

Each team member must know what the other members are working on and how their work fits into the project.

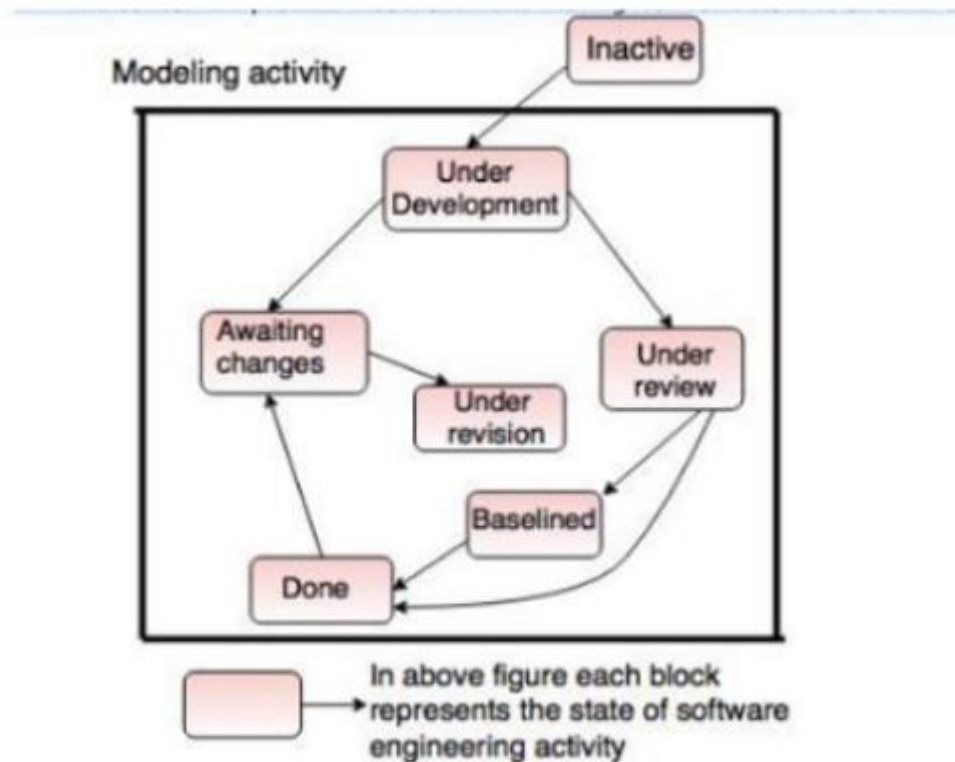


Fig. - One element of the concurrent process model

- Advantages of the concurrent development model
 - This model is applicable to all types of software development processes.
 - It is easy for understanding and use.
 - It gives immediate feedback from testing.
 - It provides an accurate picture of the current state of a project.
- Disadvantages of the concurrent development model
 - It needs better communication between the team members. This may not be achieved all the time.
 - It requires to remember the status of the different activities.

6) Spiral Model

Spiral model is a risk driven process model.

It is used for generating the software projects.

In spiral model, an alternate solution is provided if the risk is found in the risk analysis, then alternate solutions are suggested and implemented.

It is a combination of prototype and sequential model or waterfall model.

In one iteration all activities are done, for large project's the output is small.

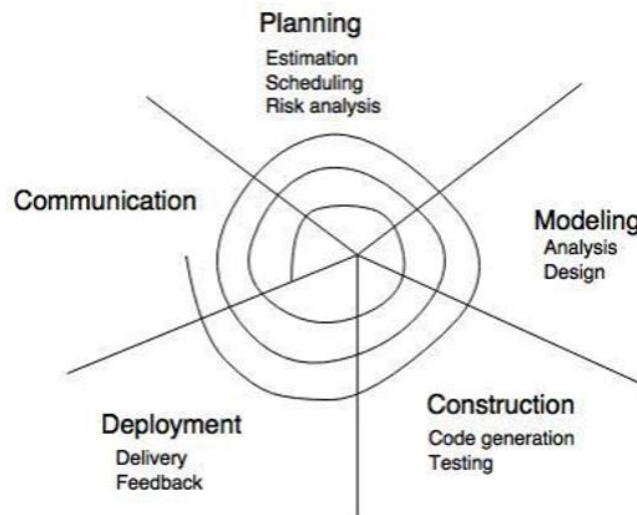


Fig. - The Spiral Model

- Advantages of Spiral Model
 - It reduces high amount of risk.
 - It is good for large and critical projects.
 - It gives strong approval and documentation control.
 - In spiral model, the software is produced early in the life cycle process.
- Disadvantages of Spiral Model
 - It can be costly to develop a software model.
 - It is not used for small projects.

7) Agile Model

The meaning of Agile is swift or versatile. "Agile process model" refers to a software development approach based on iterative development. Agile methods break tasks into smaller iterations, or parts do not directly involve long term planning. The project scope and requirements are laid down at the beginning of the development process. Plans regarding the number of iterations, the duration and the scope of each iteration are clearly defined in advance.

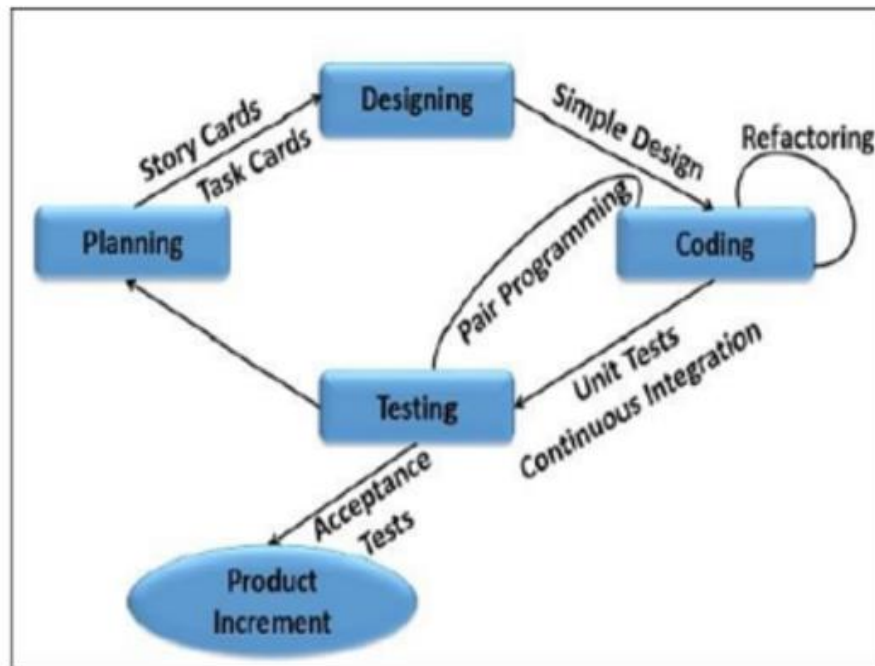


Fig. Agile Model

- Advantages:
 - Frequent Delivery
 - Face-to-Face Communication with clients.
 - Efficient design and fulfils the business requirement.
 - Anytime changes are acceptable.
 - It reduces total development time.
- Disadvantages:
 - Due to the shortage of formal documents, it creates confusion and crucial decisions taken throughout various phases can be misinterpreted at any time by different team members.
 - Due to the lack of proper documentation, once the project completes and the developers allotted to another project, maintenance of the finished project can become a difficulty.

8) Extreme Programming (XP)

XP is a lightweight, efficient, low-risk, flexible, predictable, scientific way to develop software. Extreme Programming (XP) was conceived and developed to address the specific needs of software development by small teams in the face of vague and changing requirements. Extreme Programming is one of the Agile software development methodologies. It provides values and principles to guide the team behavior. The team is expected to self-organize. Extreme Programming provides specific core practices where each practice is simple and self-complete. Combination of practices produces more complex and emergent behavior.



9) Scrum

Scrum is aimed at sustaining strong collaboration between people working on complex products, and details are being changed or added. It is based upon the systematic interactions between the three major roles: Scrum Master, Product Owner, and the Team. Scrum Master is a central figure within a project. His principal responsibility is to eliminate all the obstacles that might prevent the team from working efficiently. Product Owner, usually a customer or other stakeholder, is actively involved throughout the project, conveying the global vision of the product and providing timely feedback on the job done after every sprint. Scrum Team is a cross-functional and self-organizing group of people that is responsible for the product implementation. It should consist of up to 7 team members, in order to stay flexible and productive.

SCRUM PROCESS



7) Kanban Model

Kanban is a popular framework which is used to implement agile software development. It takes real time communication of capacity and complete transparency of work. The work items are represented in a kanban board visually, allowing team members to see the state of every piece of work at any time.

Requirement / Task / Incident Progress					
Backlog	Planned	In Progress	Developed	Tested	Completed
User Story	User Story TK TK TK	User Story	TK TK	User Story TK	User Story TK TK
User Story	IN	User Story TK	TK TK IN	TK	IN IN
User Story		IN			
User Story					
User Story					

Model we used in our IPD:

Agile Model

Our app is ML based. ML models need continuous improvements based on testing. Flexibility to add new features like disease prediction accuracy improvements or better dataset handling. Regular feedback loop (you can test with farmers/agriculture experts).

Conclusion:

Thus we successfully prepared detailed statement of problem for the selected IPD project and identify suitable process model for the same with justification.